SOKOINE UNIVERSITY OF AGRICULTURE

School of Engineering and Technology | Department of Agricultural Engineering

AEN 214: INTRODUCTION TO ELECTRONICS

OPEN COURSEWORK ASSIGNMENT 2: SPECIAL PURPOSE DIODES, BJTS & POWER FIELD EFFECT TRANSISTORS

Curriculum Coverage: Topic 4 (Special Diodes), Topic 5 (Bipolar Junction Transistors), Topic 6 (Field Effect Transistors)
Due Date / Submission: Week 9 of Semester | Individual Submission | Total Marks: 100 Marks

Instructions to Candidates:
1. This assignment is an open engineering coursework assessment designed to develop analytical and practical circuit design competence.
2. Answer ALL twelve (12) questions. Questions are tiered into Part A (Foundational Concepts), Part B (Quantitative Circuit Analysis), and Part C (Agricultural Engineering Systems Design).
3. Show all mathematical derivations, state all engineering assumptions clearly, sketch labeled schematics where applicable, and cite proper SI units.
4. Neatness, logical presentation, and clear engineering justification contribute to the final grade.

PART A: Foundational Physics & Transistor Theory [20 Marks]

Question 1 (5 Marks): Contrast Zener breakdown with Avalanche breakdown across four distinct criteria: (a) Doping level, (b) Typical breakdown voltage threshold, (c) Physical carrier multiplication mechanism, and (d) Temperature coefficient of breakdown voltage.

Question 2 (5 Marks): Explain why an NPN Bipolar Junction Transistor cannot be effectively modeled or fabricated simply by connecting two discrete back-to-back PN junction diodes with copper wire. Refer to base width, carrier lifetime, and recombination mechanics.

Question 3 (5 Marks): Describe the pinch-off phenomenon in an N-channel Junction Field Effect Transistor (JFET). Explain why the drain current saturates at a constant value (I_{DSS}) beyond pinch-off ($V_{DS} > |V_P|$) rather than dropping to zero.

Question 4 (5 Marks): Explain why power MOSFETs are immune to thermal runaway and can be operated in parallel arrays with negligible risk, whereas paralleling power BJTs without emitter ballasting resistors leads to thermal destruction.

PART B: Quantitative Transistor Circuit Analysis [35 Marks]

Question 5 (8 Marks): A 6.8V Zener diode shunt regulator is supplied from an unregulated DC input that fluctuates between 11V and 15V. The load resistance varies from 340 Ω to 1.2 kΩ. The Zener diode has ratings: $V_Z = 6.8$ V, $I_{ZK} = 3$ mA, and $P_{Z(max)} = 1.0$ W. Calculate the permissible range for series current-limiting resistor $R_{S(min)}$ and $R_{S(max)}$ using governing boundary equations and select a suitable standard E24 resistor value.

Question 6 (9 Marks): For a silicon NPN transistor configured in Common-Emitter mode with $\beta = 120$ and $I_{CBO} = 50$ nA at 25°C, (a) Calculate the Common-Base current gain $\alpha = \beta / (\beta + 1)$. (b) Calculate the open-base collector-emitter leakage current $I_{CEO} = (\beta + 1) \cdot I_{CBO}$. (c) If the junction temperature rises to 75°C causing I_{CBO} to increase eight-fold, compute the new value of I_{CEO}.

Question 7 (9 Marks): An N-channel JFET has saturation drain current $I_{DSS} = 14$ mA and pinch-off voltage $V_P = -4.5$ V. Using Shockley's equation $I_D = I_{DSS} \cdot [\ 1 - (V_{GS} / V_P)\ ]^2$: (a) Calculate drain current I_D for $V_{GS} =$ 0V, -1.0V, -2.0V, -3.0V, and -4.5V. (b) Calculate maximum transconductance $g_{m0} = 2 \cdot I_{DSS} / |V_P|$. (c) Compute operating transconductance g_m at $V_{GS} = -2.0$ V.

Question 8 (9 Marks): A Common-Emitter BJT amplifier with voltage-divider bias has circuit values: $V_{CC} =$ 15 V, $R_1 = 47$ kΩ, $R_2 = 12$ kΩ, $R_C = 3.3$ kΩ, $R_E = 1.2$ kΩ, and $\beta = 100$. (a) Determine the DC quiescent operating point (I_{CQ}, V_{CEQ}). (b) Calculate the small-signal dynamic emitter resistance $r_e = 26$ mV / I_{EQ}. (c) Calculate the AC voltage gain $A_v = -(R_C \parallel R_L) / r_e$ assuming the emitter resistor is bypassed with a large capacitor C_E.

PART C: Agricultural Actuator & Power Electronics Design [45 Marks]

Question 9 (11 Marks) - Solar Array Shading Protection:
A solar-powered borehole water pumping installation comprises three series strings of 400W monocrystalline photovoltaic modules. Each module produces a string current $I_{string} = 10.5$ A. During dusty dry-season conditions in central Tanzania, acacia tree leaves shade one substring.
(a) Explain the electrical and thermal consequences of cell reverse breakdown if bypass diodes are omitted.
(b) Compare the power dissipation in a sealed IP67 junction box when using a standard silicon rectifier diode ($V_F = 1.0$ V at 10.5 A) versus an MBR1645 Schottky barrier diode ($V_F = 0.42$ V at 10.5 A) using $P = I \cdot V_F$.
(c) If the junction box has a thermal resistance to ambient $\theta_{ca} = 8.5$ °C/W and ambient temperature is 42°C, compute the steady-state temperature inside the enclosure $T_{case} = T_{amb} + P \cdot \theta_{ca}$ for both diode types.

Question 10 (11 Marks) - BJT Saturated Switch for Grain Aeration Fan:
An automated maize storage silo uses an ESP32 microcontroller (3.3V logic, maximum safe GPIO current = 12 mA) to actuate a 12V, 15W cooling fan relay.
(a) Calculate the required collector saturation current $I_{C(sat)} = P / V_{CC}$.
(b) The relay is switched using a BD139 NPN power transistor with minimum saturation gain $\beta_{min} = 40$. Using an overdrive factor ODF = 3.0 to ensure solid saturation across temperature extremes, compute the required base current $I_{B(forced)} = ODF \cdot [\ I_{C(sat)} / \beta_{min}\ ]$.
(c) Sizing the series base resistor $R_B = (V_{GPIO} - V_{BE(sat)}) / I_{B(forced)}$ (assume $V_{BE(sat)} = 0.85$ V) and check whether the ESP32 GPIO pin current rating is respected.
(d) Explain the necessity of the anti-parallel freewheeling diode across the relay coil and specify an appropriate diode part number.

Question 11 (11 Marks) - Power MOSFET Thermal Sizing for 24V Solar Irrigation Pump:
A 24V, 360W DC submersible solar pump (full-load current $I_D = 15$ A) is speed-controlled via PWM from an agricultural automation board. An IRF3205 N-channel Power MOSFET is selected, having on-state resistance $R_{DS(on)} = 8.0$ mΩ (0.008 Ω), junction-to-case thermal resistance $\theta_{jc} = 0.75$ °C/W, and maximum rated junction temperature $T_{j(max)} = 175$°C. The control box operates in an outdoor shed where ambient temperature reaches $T_{amb} = 45$°C.
(a) Calculate the static conduction power loss $P_{cond} = I_D^2 \cdot R_{DS(on)}$ in the MOSFET at full 15A continuous current.

(b) If junction-to-ambient thermal resistance without a heatsink is θ_{ja} = 62 °C/W, prove that the MOSFET will burn out if operated without a heatsink via $T_j = T_{amb} + P_{cond} \cdot \theta_{ja}$.
(c) Calculate the maximum allowable heatsink thermal resistance $\theta_{sa} = [(T_j - T_{amb}) / P_{cond}] - \theta_{jc} - \theta_{cs}$ to maintain the MOSFET junction temperature below a safe design limit of T_j = 90°C (assume thermal grease interface θ_{cs} = 0.50 °C/W).

Question 12 (12 Marks) - High-Impedance Soil pH Transducer Buffer:
Glass-membrane electrochemical soil pH sensors have an extraordinarily high internal source resistance (R_{source} ≈ 100 MΩ to 1 GΩ) and output an analog signal of approximately 59.2 mV per pH unit at 25°C. Connecting this probe to a standard BJT amplifier causes severe signal attenuation.
(a) Explain why a standard BJT amplifier cannot be used for direct buffering of a glass pH electrode.
(b) Design a Common-Drain (Source Follower) buffer amplifier using a 2N5457 N-channel JFET (I_{DSS} = 5.0 mA, V_P = -2.5 V). Draw the circuit schematic and derive the expression for input resistance R_{in} and voltage gain $A_v = (g_m \cdot R_S) / (1 + g_m \cdot R_S)$.
(c) What precautions regarding PCB contamination (humidity, agricultural fertilizer dust, and flux residue) must be taken to maintain 10^{12} Ω insulation resistance on the sensor board?